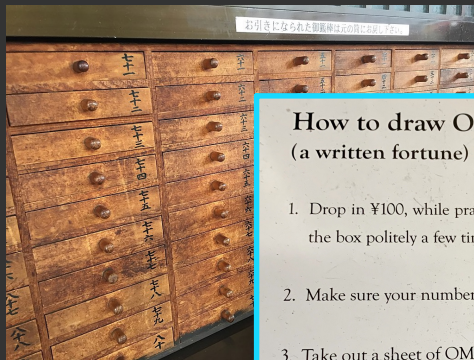


probability

CSCI
270

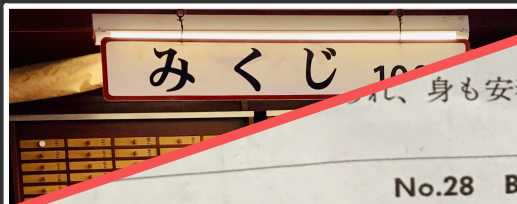
omikuji



How to draw OMIKUJI (a written fortune)

1. Drop in ¥100, while praying for your wish, shake the box politely a few times.
2. Make sure your number and put the stick back.
3. Take out a sheet of OMIKUJI from the drawer of your number.
4. When you draw a good fortune, please take it home. But you should not be careless and arrogant.
5. When you draw a bad fortune, please do not worry. Tie it on the hanger and drop bad fortune off here.

omikuji



新築・引越：悪いでしょう。

結婚・付き合い：悪いでしょう。

う。

No.28 BAD FORTUNE

Although you are in a hurry, there will be no boat to cross the river. If you dare to sail, the boat will be swallowed by high waves.

You can't go ahead and should go back to your homeland. When you get home, you will not meet any crisis and your mind will be at peace.

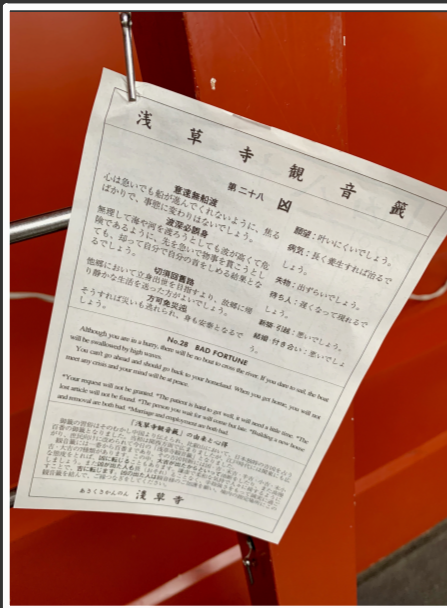
*Your request will not be granted. *The patient is hard to get well, it will need a little time. *The lost article will not be found. *The person you wait for will come but late. *Building a new house and removal are both bad. *Marriage and employment are both bad.

「浅草寺観音籤」の由来と心得

御籤の習俗はそのむかし中国より伝えられ、比叡山において、日本独特の吉凶判断の
百番の御籤となりました。当初は関西方面で広まりましたが、江戸時代には全国的に
がり、庶民向けに改められて今日の「浅草寺観音籤」となりました。
観音籤には一番から百番まであり、その吉凶判断には凶・吉
吉・大吉の7種類があります。この中、大吉が出たからと
な態度をとれば、凶に転じることもあります。
なましよう。また凶が出た人も、
すことで、吉に転じます。

come, please do not
the hanger and drop bad fortune
off here.

omikuji



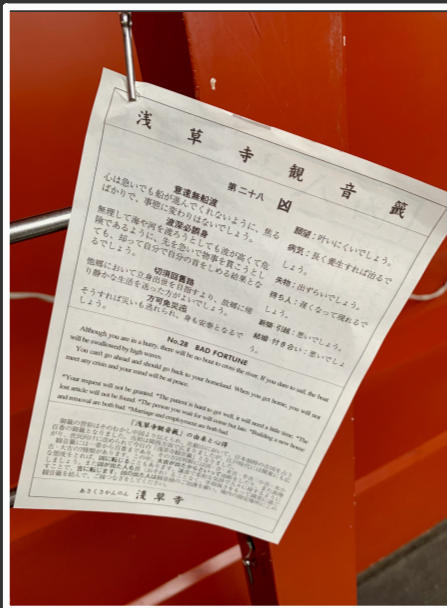
Fortunes [\[edit \]](#)

The standard *Ganzan Daishi Hyakusen* sequence contains the following fortunes (from best to worst):

- Great blessing (大吉, *dai-kichi*)
- Blessing (吉, *kichi*)
- Small blessing (小吉, *shō-kichi*)
- Half-blessing (半吉, *han-kichi*)
- Future blessing (末吉, *sue-kichi*)
- Future small blessing (末小吉, *sue-shō-kichi*)
- Misfortune (凶, *kyō*)

from the wikipedia article "o-mikuji"

omikuji



Fortunes [\[edit \]](#)

The standard *Ganzan Daishi Hyakusen* sequence contains the following fortunes (from best to worst):

- Great blessing (大吉, *dai-kichi*)
- Blessing (吉, *kichi*)
- Small blessing (小吉, *shō-kichi*)
- ~~Half blessing (半吉, *han-kichi*)~~
- ~~Future blessing (末吉, *sue-kichi*)~~
- ~~Future small blessing (末小吉, *sue shō kichi*)~~
- Misfortune (凶, *kyō*)

from the wikipedia article "o-mikuji"



great blessing

small misfortune

red : blessing

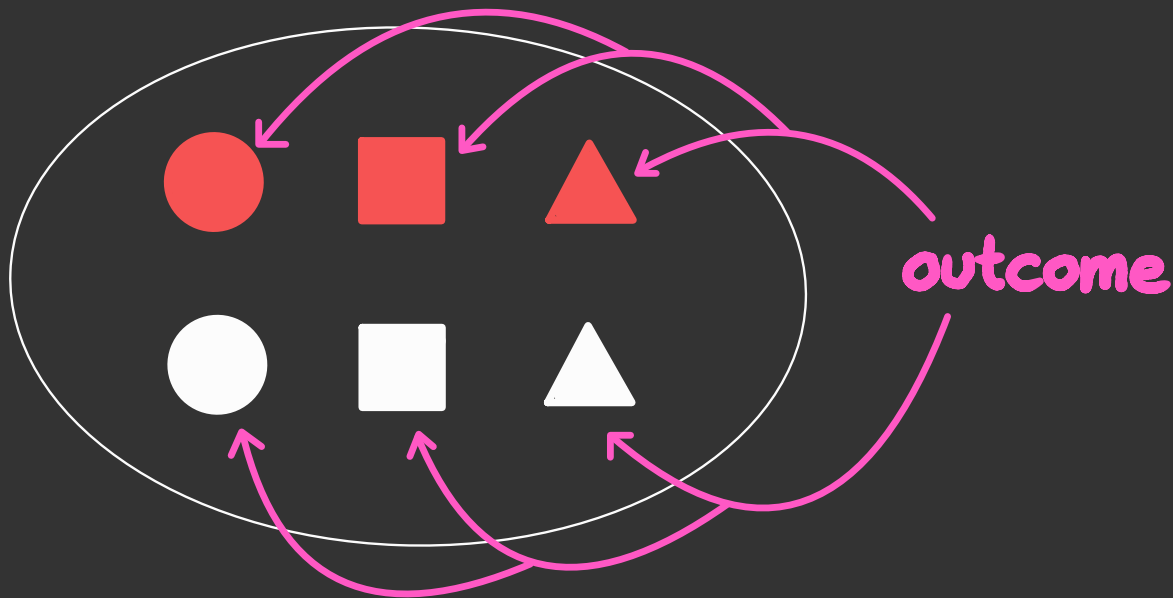
white : misfortune

how do we model omikuji using

probability ?

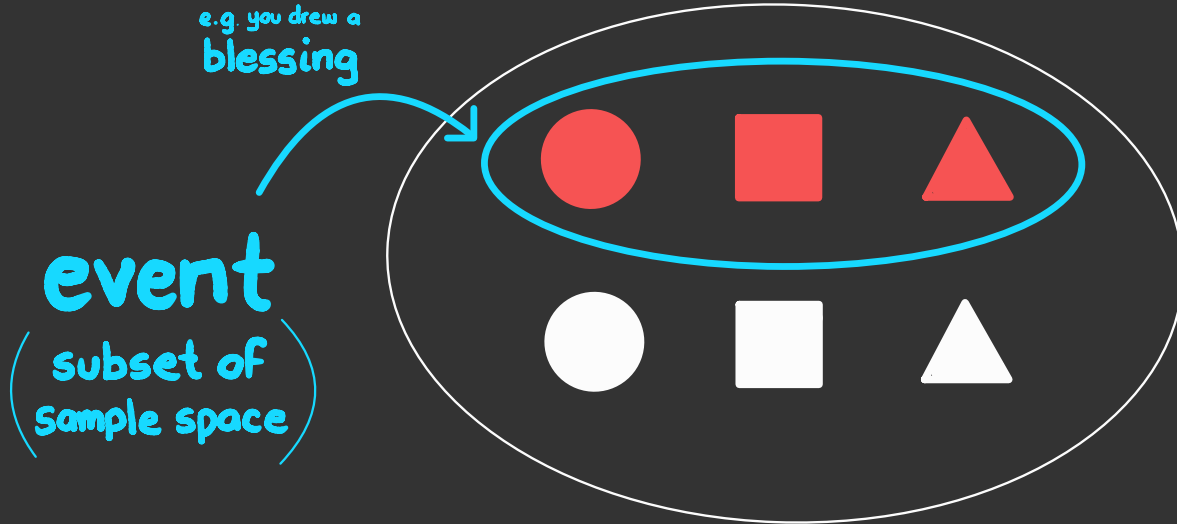
sample space

(set of possible outcomes)



sample space

(set of possible outcomes)



probability

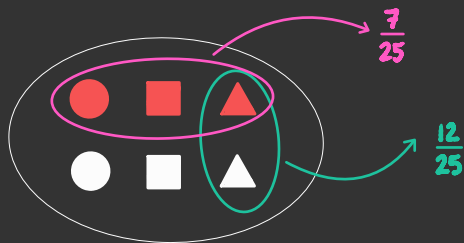
a probability law is a function P that maps each event to a nonnegative number

such that:

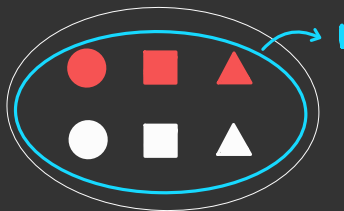
$$P(\Omega) = 1$$

if A and B are disjoint events,

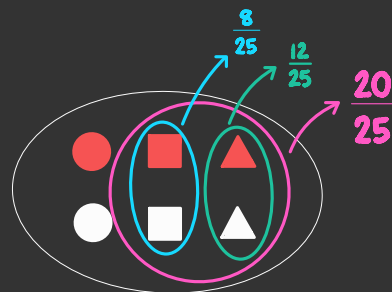
$$P(A \cup B) = P(A) + P(B)$$



sample space Ω
(set of possible outcomes)



sample space Ω
(set of possible outcomes)



sample space Ω
(set of possible outcomes)

a probability law is a function P that maps each event to a nonnegative number

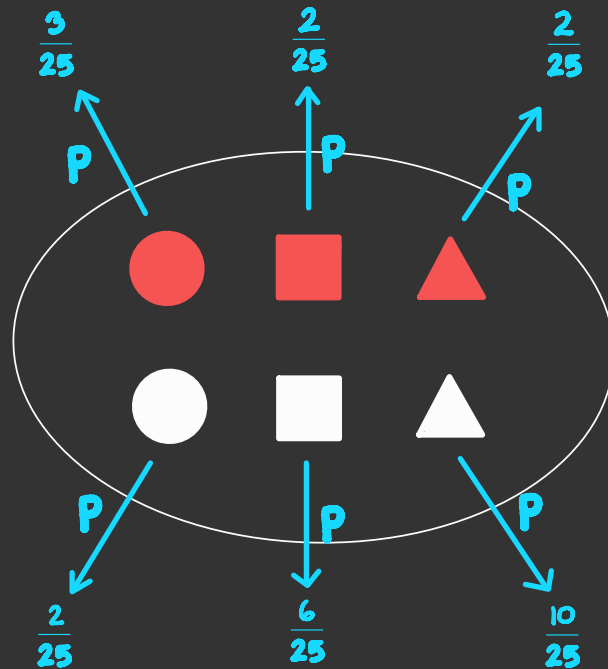


△	○	●	△	□
■	□	△	□	△
●	□	●	△	□
△	△	□	△	△
△	△	■	△	○

such that:

$$P(\Omega) = 1$$

if A and B are disjoint events,
 $P(A \cup B) = P(A) + P(B)$



a probability law is a function P
that maps each event to a
nonnegative number

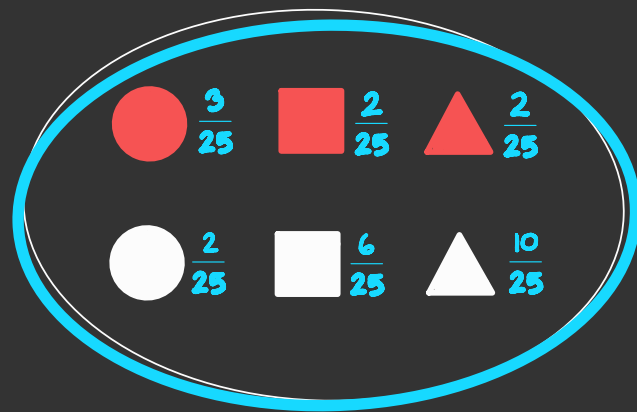


▲	●	●	▲	□
■	□	▲	□	▲
●	□	●	▲	□
▲	▲	□	▲	▲
▲	▲	■	▲	●

such that:

$$P(\Omega) = 1$$

if A and B are disjoint events,
 $P(A \cup B) = P(A) + P(B)$



$$\frac{3}{25} + \frac{2}{25} + \frac{2}{25} + \frac{2}{25} + \frac{6}{25} + \frac{10}{25} = 1$$

a probability law is a function P
that maps each event to a
nonnegative number

such that:

$$P(\Omega) = 1$$

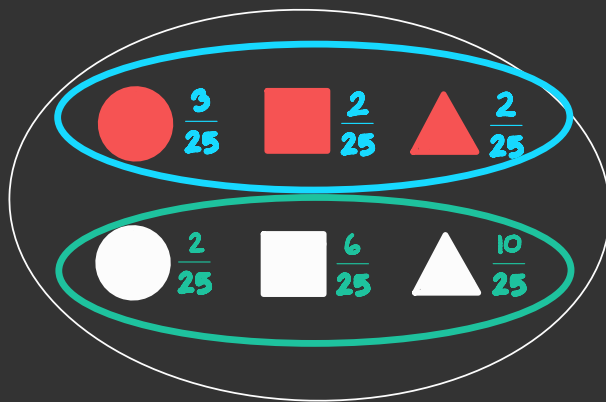
if A and B are disjoint events,

$$P(A \cup B) = P(A) + P(B)$$



▲	●	●	▲	□
■	□	▲	□	▲
●	□	●	▲	□
▲	▲	□	▲	▲
▲	▲	■	▲	●

$$P(\text{you draw a blessing}) = \frac{3}{25} + \frac{2}{25} + \frac{2}{25}$$



$$P(\text{you draw a misfortune}) = \frac{2}{25} + \frac{6}{25} + \frac{10}{25}$$